Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 1

Semester Test 1

Module EBN111

Elektrisiteit en Elektronika

15 Maart 2012

Module EBN111

Electricity and Electronics

15 March 2012

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:* Memo.			
Maksimum punte: *Maximum marks:*	45	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 minute *90 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total numer of pages (including this page):*		13	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
   All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
   Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.
   Final answers must be written as real numbers, and not as fractions.

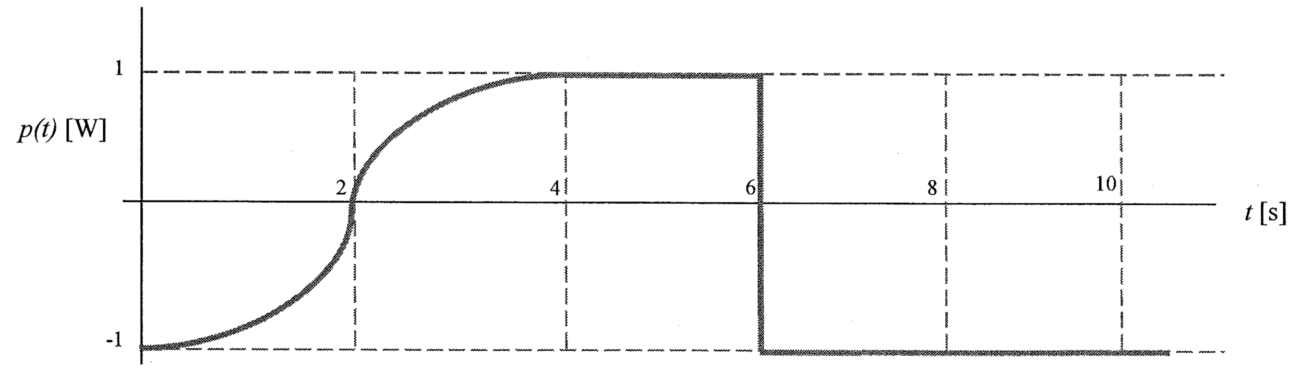
Dosent(e): *Lecturer(s):*	1 J Joubert
	2 JD le Roux

Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding.
Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.

Groephoof: *Group Head:*	Prof J Joubert	Handtekening: *Signature:*	

VRAAG/QUESTION 1 [22]

Vraag 1.1 [2]	**Question 1.1 [2]**
Die grafiek dui die drywing van `n element teenoor tyd aan. Bepaal die hoeveelheid energy wat die element verbruik tussen 4 en 10 sekondes.	The graph shows the power of an element over time. Determine the energy used by the element between 4 and 10 seconds.



$$p = \frac{dw}{dt}$$

$$\therefore W = \int_{t_0}^{t_1} p \, dt$$

$$= (6-4) \times 1 + (10-6) \times -1$$

$$= 2 - 4$$

$$= -2 \text{ J} \quad \checkmark\checkmark$$

(2 marks for correct sign (-) and size (2))

(0 marks for any other answer.)

Vraag 1.2 [2]	**Question 1.2 [2]**
Dit verg 600 mJ om -60 nC in 6 μs deur 'n element te beweeg. Bepaal die spanning oor die element en die stroom deur die element. (Skryf die antwoorde deur gebruik te maak van die SI voorvoegsels: M, k, m, μ, n, of p.)	To move -60 nC in 6 μs through an element requires 600 mJ. Determine the voltage across the element and the current through the element. (Write the answers using the SI prefixes: M, k, m, μ, n, or p.)

* $i = \Delta q / \Delta t$

$= \dfrac{-60\,nC}{6\,\mu s}$

$= \underline{-10\,mA}$ ✓

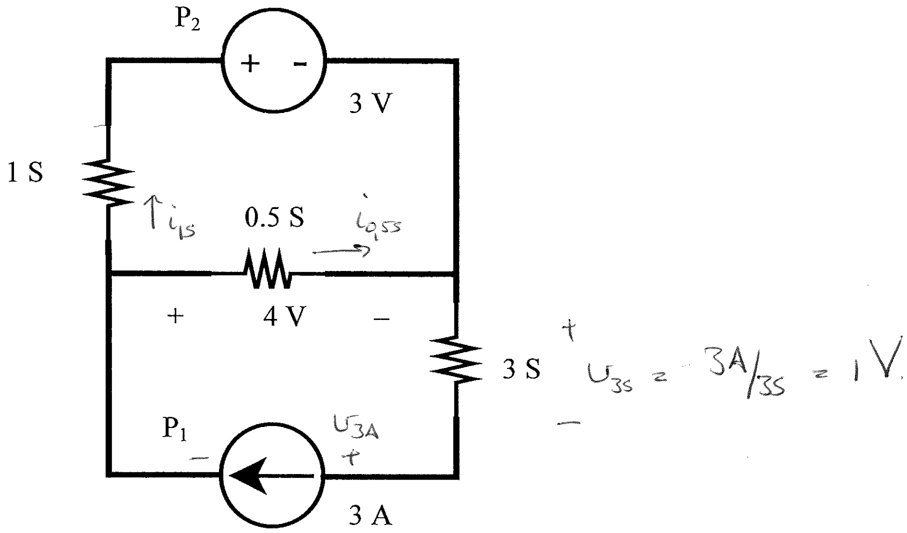
* $v = dw/dq$

$= \dfrac{600\,mJ}{-60\,nC}$

$= \underline{-10\,MV}$ ✓

(1 mark if answer has
- correct sign
- correct size
- SI prefix)

Vraag 1.3 [3] a) Bepaal die drywing P_1 en P_2. b) Vir P_2, word drywing geabsorbeer of gelewer?	**Question 1.3 [3]** a) Determine the power P_1 and P_2. b) For P_2, is power absorbed or provided?



* $i_{0.5S} = 0.5 \times 4 = 2A.$

* $i_{1S} = 3 - 2 = 1\,A$

* $P_2 = 3V \times 1A = 3W$ ✓

KVL: $4V + \frac{3A}{3S} + U_{3A} = 0$

$U_{3A} = -5V$

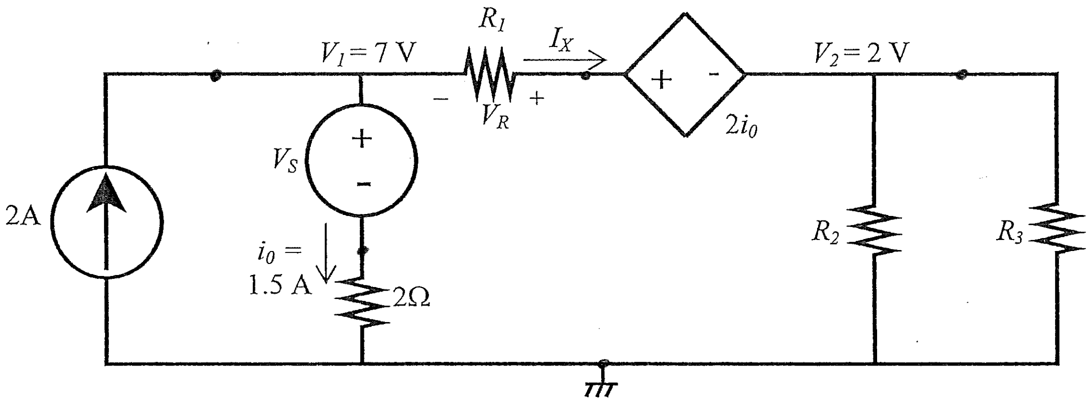
* $P_1 = -5V \times 3A = -15\,W.$ ✓

→ P_2 absorbs power. ✓ (Marked according to value given for P_2.)

→ If answer of P_2 was given as negative, then student must write: P_2 delivers power.

Vraag 1.4 [7]
Weerstande R_1, R_2 en R_3 en bronspanning V_S is onbekend.
(a) Hoeveel nodusse is daar in die stroombaan? [1]
(b) Hoeveel takke is daar in die stroombaan? [1]
(c) Bepaal die spanning V_S.[1]
(d) Bepaal die spanning V_R oor weerstand R_1. [2]
(e) Bepaal stroom I_X.[2]

Question 1.4 [7]
Source voltage V_S and resistors R_1, R_2 and R_3 are unknown.
(a) How many nodes are there in the circuit? [1]
(b) How many branches are there in the circuit? [1]
(c) Determine the voltage V_S. [1]
(d) Determine the voltage V_R over resistor R_1. [2]
(e) Determine the current I_X. [2]



a) 5 nodes ✓ b) 7 branches. ✓

Remember $b = n + (\text{independent loops}) - 1$

$7 = 5 + 3 - 1.$

c) KVL: $-V_1 + V_s + i_o \times 2 = 0$

$-7 + V_s + 3 = 0$

$V_s = 4V$ ✓

d) KVL: $-V_1 + 2i_o + V_2 - V_R = 0$

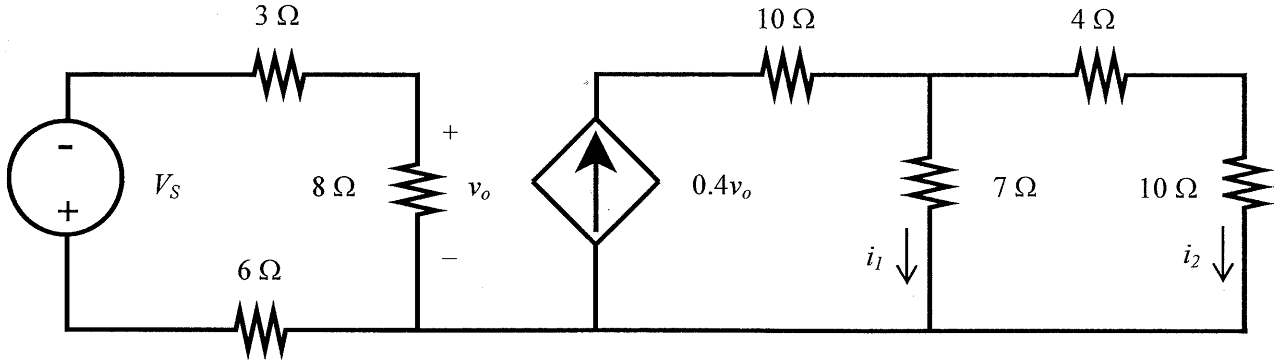
$V_R = -7 + 2 + 2(1{,}5)$

$= -2V$ ✓✓

(If size correct, but sign wrong, 1 mark)

e) KCL: $I_x = 2 - 1{,}5 = 0{,}5\,A$ ✓✓

Vraag 1.5 [4]	Question 1.5 [4]
a) Bereken v_0 indien $V_S = 21.25$ V.	a) Determine v_0 if $V_S = 21.25$ V.
b) Bereken i_2 indien $v_0 = 5$ V.	b) Determine i_2 if $v_0 = 5$ V.

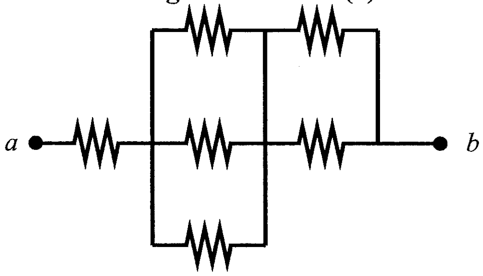
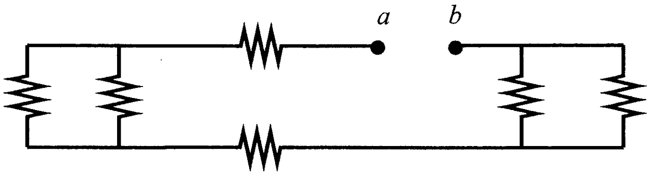


a) $v_o = \dfrac{-V_s}{6+3+8} \times 8$

$= -10V$ ✓✓ (If size is correct but sign is wrong, then one mark.)

b) $i_2 = \dfrac{0,4(5) \times 7}{4+10+7}$

$= 0,6667\ A$ ✓✓ (Two marks for right answer.)

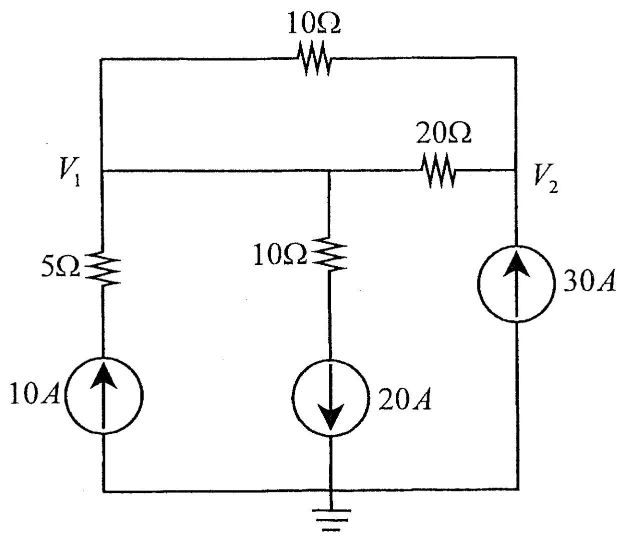
Vraag 1.6 [4] Bepaal die ekwivalente weerstand by terminale *a-b* vir kringbaan (1) en (2). Alle resistors is 2 Ω.	**Question 1.6 [4]** Determine the equivalent resistance at terminals *a-b* for circuit (a) and (b). All resistors are 2 Ω.
Kringbaan/*Circuit* (1)  	**Kringbaan/*Circuit* (2)** 

$$R_{eq} = 2 + 2\|2\|2 + 2\|2$$
$$= 2 + 2/3 + 1$$
$$= \underline{3{,}667\,\Omega}$$

$$R_{eq} = 2 + 2\|2 + 2 + 2\|2$$
$$= 4 + 1 + 1$$
$$= \underline{6\,\Omega}$$

VRAAG/QUESTION 2 [23]

Vraag 2.1[4]	Question 2.1[4]
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $$aV_1 + bV_2 = -200$$ $$cV_1 + dV_2 = 600$$	Use node analysis to set up two equations of the following form for the circuit: $$aV_1 + bV_2 = -200$$ $$cV_1 + dV_2 = 600$$



Node 1: $-10 + 20 + \frac{V_1 - V_2}{20} + \frac{V_1 - V_2}{10} = 0$

$\times 20$: $200 + V_1 - V_2 + 2V_1 - 2V_2 = 0$

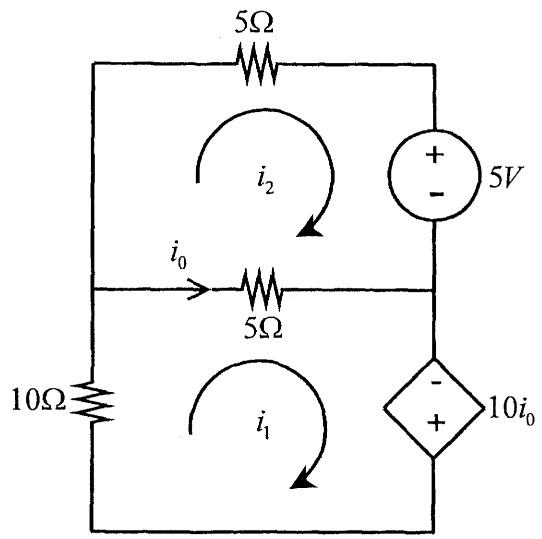
$$3V_1 - 3V_2 = -200$$

Node 2: $-30 + \frac{V_2 - V_1}{20} + \frac{V_2 - V_1}{10} = 0$

$\times 20$: $-600 + V_2 - V_1 + 2V_2 - 2V_1 = 0$

$$-3V_1 + 3V_2 = 600$$

Vraag 2.2[4]	Question 2.2 [4]
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $$ai_1 + bi_2 = 0$$ $$ci_1 + di_2 = -5$$	Use mesh analysis to set up two equations of the following form for the circuit: $$ai_1 + bi_2 = 0$$ $$ci_1 + di_2 = -5$$



mesh 1: $10i_1 + 5(i_1 - i_2) - 10i_0 = 0$

but $i_0 = i_1 - i_2$

$$\cancel{10i_1} + 5i_1 - 5i_2 - \cancel{10i_1} + 10i_2 = 0$$

$$5i_1 + 5i_2 = 0$$

mesh 2: $5i_2 + 5 + 5(i_2 - i_1) = 0$

$$-5i_1 + 10i_2 = -5$$

Vraag 2.3 [4]

(a) Bepaal V_1.

$$\begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

(b) Bepaal i_2.

$$\begin{bmatrix} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ -10 \end{bmatrix}$$

Question 2.3 [4]

(a) Determine V_1.

$$\begin{bmatrix} 5 & -3 \\ -3 & 6 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

(b) Determine i_2.

$$\begin{bmatrix} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & 10 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ -10 \end{bmatrix}$$

(a) $$\frac{\begin{vmatrix} -1 & -3 \\ 2 & 6 \end{vmatrix}}{\begin{vmatrix} 5 & -3 \\ -3 & 6 \end{vmatrix}} = \frac{-6+6}{30-9} = \frac{0}{21} = 0 \longrightarrow$$

(b) $$\Delta = \begin{array}{|ccc|} 5 & -5 & 0 \\ -2 & 20 & 0 \\ 0 & 0 & 10 \end{array} = (5 \times 20 \times 10) - ((-2) \times (-5) \times 10)$$

$$\begin{array}{ccc} 5 & -5 & 0 \\ -2 & 20 & 0 \end{array}$$

$$= 1000 - 100 = 900$$

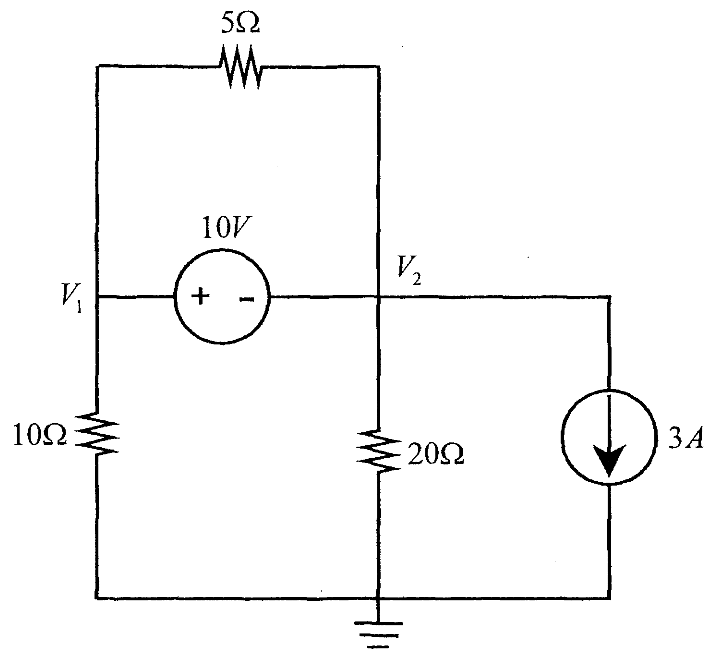
$$\Delta_2 = \begin{array}{|ccc|} 5 & 20 & 0 \\ -2 & 10 & 0 \\ 0 & -10 & 10 \end{array} = (5 \times 10 \times 10) - (-2 \times 20 \times 10)$$

$$\begin{array}{ccc} 5 & 20 & 0 \\ -2 & 10 & 0 \end{array}$$

$$= 500 + 400 = 900$$

$$i_2 = \frac{\Delta_2}{\Delta} = \frac{900}{900} = 1 \longrightarrow$$

Vraag 2.4 [4]	**Question 2.4 [4]**
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $aV_1 + bV_2 = 10$ $cV_1 + dV_2 = -60$	Use node analysis to set up two equations of the following form for the circuit: $aV_1 + bV_2 = 10$ $cV_1 + dV_2 = -60$



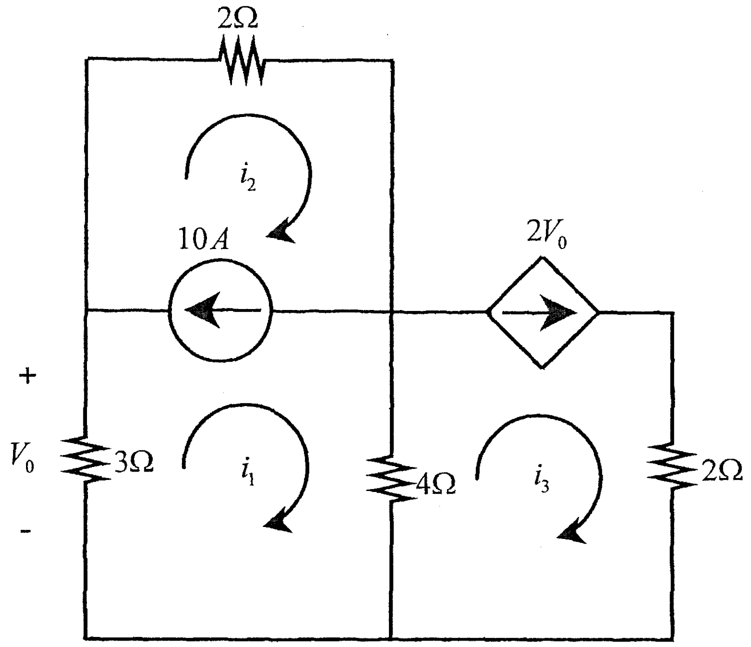
Supernode: $V_2 + 10 = V_1$

$V_1 - V_2 = 10$

KCL: $\frac{V_1}{10} + \frac{V_2}{20} + 3 = 0$

×20: $2V_1 + V_2 = -60$

Vraag 2.5 [4]	**Question 2.5 [4]**
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel vir die stroombaan: $$ai_1 + bi_2 = 10$$ $$ci_1 + di_2 = 0$$	Use mesh analysis to set up two equations of the following form for the circuit: $$ai_1 + bi_2 = 10$$ $$ci_1 + di_2 = 0$$



$i_3 = 2V_0 = 2(-3i_1) = -6i_1$

KVL: $3i_1 + 2i_2 + 4(i_1 - i_3) = 0$

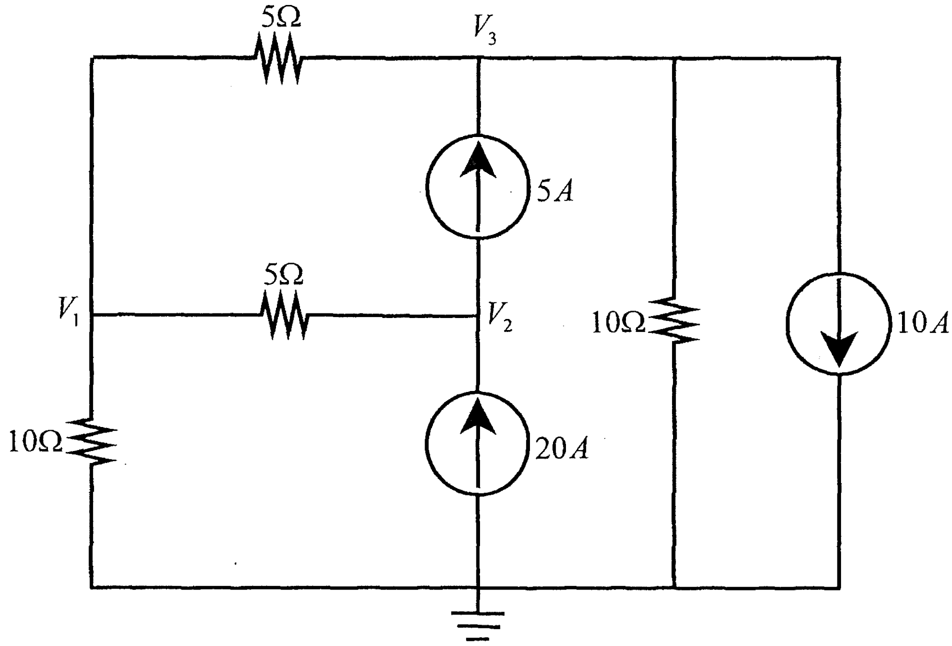
$3i_1 + 2i_2 + 4i_1 - 4(-6i_1) = 0$

$31i_1 + 2i_2 = 0$

Supermesh: $i_2 - i_1 = 10$

$-i_1 + i_2 = 10$

Vraag 2.6 [3]	Question 2.6 [3]
Bepaal die node-spannings vergelykings deur inspeksie.	Determine the node-voltage equations by inspection.



$$\begin{bmatrix} 0.5 & -0.2 & -0.2 \\ -0.2 & 0.2 & 0 \\ -0.2 & 0 & 0.3 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 15 \\ -5 \end{bmatrix}$$

✓ for diagonal self terms
✓ for off-diagonal coupling terms
✓ for right hand excitation matrix